UNIVERSITY OF CAMBRIDGE | PHD RESEARCH PROPOSAL

From Retrieval to Commitment

Selective and Revisable Claim Verification under Evolving Evidence

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Summary

Automated fact-checking is usually evaluated as a one-shot pipeline: retrieve evidence for a claim, predict a verdict, and generate a justification. This abstraction is inadequate for open-world verification. Evidence may initially be insufficient, ambiguous, duplicated across dependent sources, temporally inapplicable, corrected, retracted, or superseded by later observations. A reliable verifier must therefore decide not only what verdict is most likely, but whether the available evidence warrants a verdict now, how that decision should change as evidence evolves, and how the resulting state should constrain what the system communicates.

This PhD will study claim verification as a sequential, risk-sensitive commitment process over evolving evidence. The central thesis is that trustworthy verification requires three separately modelled and evaluated capabilities: (i) calibrated commitment - continuing evidence acquisition, deferring, or issuing a verdict under an explicit risk criterion; (ii) revisable evidence state - maintaining temporal scope, provenance, dependencies, and revision lineage as evidence changes; and (iii) state-faithful communication - producing verdicts and explanations that use active evidence, expose residual uncertainty, and do not reintroduce stale or rejected material. The research will develop a longitudinal task formulation, methods for selective evidence acquisition and revision, and an evaluation framework capable of locating failures at each layer.

Background and research gap

Modern automated fact-checking has progressed from closed-source claim classification to evidence-grounded verification. FEVER established joint evidence retrieval and verdict prediction; AVeriTeC introduced real-world claims, open-web evidence, intermediate questions, and temporally controlled annotation; and Ev2R evaluates whether retrieved evidence both matches references and supports a verdict (Thorne et al., 2018; Schlichtkrull et al., 2023; Akhtar et al., 2026). Recent work has addressed important limitations of the one-shot formulation. AmbiFC and Dis2Dis model ambiguous claims and

their interpretations; Fact Checking with Insufficient Evidence asks systems not to decide without adequate support; FIRE performs iterative retrieval rather than using a fixed evidence budget; TSVer studies temporal and numerical evidence; and PledgeTracker organises incremental evidence into event timelines (Atanasova et al., 2022; Glockner et al., 2024; Staliunaite and Vlachos, 2025; Xie et al., 2025; Strong and Vlachos, 2025; Chen et al., 2025).

These advances cover complementary parts of a larger unresolved problem. Existing tasks generally evaluate either a static evidence snapshot, an iterative search episode ending in one answer, a domain-specific timeline, or the periodic refresh of a benchmark. They do not ordinarily evaluate how a verifier should maintain and revise a provenance-rich state for the same claim across successive evidence snapshots, while controlling false commitment and ensuring that later explanations remain consistent with the revised state. Forecast-updating and temporally valid agent-evaluation work demonstrate the importance of measuring belief change and time-bounded evidence, but concern probabilistic future events rather than evidence-supported verdicts about claims (Yuan et al., 2025; Chi et al., 2026). Classical belief-revision and truth-maintenance systems offer useful principles, including minimal change, dependency tracking, and reason maintenance, but assume symbolic propositions and do not resolve the uncertainty, source dependence, retrieval cost, and linguistic ambiguity of open-web evidence (Alchourron et al., 1985; Doyle, 1979).

The proposed novelty is deliberately scoped. It is not the invention of abstention, temporal fact-checking, iterative retrieval, belief revision, provenance tracking, or dynamic benchmarks. It is a unified empirical formulation of selective commitment, longitudinal evidence-state revision, and state-constrained communication, with transition-level supervision and controlled evaluation connecting these layers.

Central thesis and objectives

For a claim observed at time t, the verifier will maintain a state:

$$S_t = (c, I_t, E_t, p_t, u_t, tau_t, L_t).$$

Here c is a normalised proposition; I_t contains viable interpretations and scope conditions; E_t is an evidence graph with sources, provenance, publication and valid times, stance, and dependency links; p_t is a verdict distribution; u_t represents evidence sufficiency and uncertainty; tau_t is the interval for which the state is asserted to hold; and L_t records revision lineage. The operational action space will include evidence acquisition, deferral, commitment to a verdict, and revision of a prior commitment. This is a task model rather than a claim that a language model possesses human beliefs.

The objectives are to:

1. Define and validate a longitudinal verification task in which claims, evidence, and verdict commitments evolve.

2. Develop calibrated policies that decide when to search further, defer, commit, or revise under accuracy-coverage-cost constraints.
3. Identify state representations and update rules that are responsive to decisive evidence but stable under irrelevant, stale, or duplicated material.
4. Determine whether compiling the maintained state into constraints for generation reduces unsupported, stale, and premise-following outputs.

Research questions

RQ1 - Selective commitment. Which observable signals of evidence sufficiency, ambiguity, temporal validity, source independence, and predictive uncertainty support a calibrated decision to continue searching, defer, or commit, and how do these policies trade off error, coverage, and retrieval cost?

RQ2 - Longitudinal revision. Which state representation and update rules best preserve provenance and produce appropriately stable yet responsive verdict trajectories when evidence is added, corrected, contradicted, duplicated, withdrawn, or rendered temporally inapplicable?

RQ3 - State-faithful communication. To what extent does constraining verdict explanations and downstream answers by the maintained evidence state reduce unsupported claims, stale-evidence use, and compliance with false premises without making outputs uninformative?

Methodology

Work package 1: task formulation and longitudinal testbed

I will first construct a reproducible testbed of approximately 300-500 claim trajectories, each containing three to five temporally ordered evidence snapshots. The textual and time-series core will draw on existing real-world resources such as AVeriTeC and TSVer, public ClaimReview records, official-data releases, corrections and retractions, and archived web pages. The aim is not to crawl the live web at unrestricted scale, but to create fixed, timestamped evidence collections that support repeatable experiments. A live-web track may later test robustness, but will not be required for the thesis.

Trajectories will represent real transitions - new corroboration, decisive contradiction, source correction, updated official statistics, temporal expiry, and newly available evidence - and controlled interventions that isolate causal effects, such as adding irrelevant passages, duplicated reporting, stale evidence, or a conflicting low-authority source. Annotation will cover claim normalisation and interpretation, evidence relevance and sufficiency, source and dependency provenance, valid time, verdict distribution, and the appropriate transition. A pilot will refine the taxonomy; the main set will use independent double annotation, adjudication, agreement reporting, and a trajectory-level train/development/test split that prevents future snapshots or near-duplicate claims from leaking across partitions.

The first contribution will be the task definition, data model, annotation study, and diagnostic evaluation showing which apparently similar systems fail at commitment, state maintenance, or communication.

Work package 2: calibrated sequential evidence commitment

The second work package will develop policies over actions such as SEARCH(query), DEFER, COMMIT(verdict), and REVISE(verdict). I will begin with interpretable and falsifiable methods: supervised action prediction, uncertainty and evidence-sufficiency thresholds, cost-sensitive decision rules, and conformal risk control where its exchangeability and monotonicity assumptions are defensible (Angelopoulos et al., 2024). Reinforcement learning or bandit optimisation will be introduced only if sequential credit assignment provides a measurable advantage over these simpler alternatives.

Candidate signals will include entailment and contradiction strength, evidence coverage, agreement across independent sources, temporal fit, interpretation entropy, retrieval redundancy, calibration estimates, and the expected marginal value of another query. The policy will be evaluated against fixed-budget retrieval, static AVeriTeC-style pipelines, confidence-threshold abstention, Fact Checking with Insufficient Evidence, FIRE, and adaptive retrieval or generation approaches. The scientific question is not whether more searching improves accuracy, but whether a system can acquire enough of the right evidence while meeting a specified false-commitment risk at useful coverage and cost.

Work package 3: revisable evidence states and faithful outputs

The third work package will compare representations for maintaining evidence and verdict dependencies across time. The proposed representation is a compact provenance graph or ledger containing atomic evidence units, source and temporal metadata, support or attack relations, dependency clusters, active, contested, or retired status, and links explaining each verdict revision. Update methods will be compared with stateless recomputation at every snapshot, latest-evidence or overwrite policies, source-weighted voting, Bayesian or log-odds aggregation, temporal knowledge-graph baselines, and truth-maintenance-inspired dependency rules.

The maintained state will then be compiled into an answer contract: outputs must identify the current verdict and its time scope; cite active supporting and conflicting evidence; distinguish unresolved ambiguity from lack of evidence; exclude retired evidence unless explaining a revision; and resist false premises in the query. Controlled ablations will separate gains due to retrieval, commitment policy, state representation, and generation constraints. Negative results will remain informative: the project explicitly tests whether accurate state maintenance is sufficient to improve natural-language behaviour rather than assuming that it is.

Evaluation

Evaluation will follow the scientific decomposition rather than collapse performance into one score. At the commitment layer, I will report selective risk-coverage curves, false-commitment and unnecessary-

deferral rates, calibration (Brier score, log loss, and calibration error), evidence-sufficiency accuracy, and retrieval cost measured in queries, sources, and tokens. At the state layer, metrics will include transition accuracy, responsiveness to decisive evidence, stability under irrelevant or duplicated evidence, temporal validity, provenance and lineage fidelity, and stale-active or contradiction rates. At the communication layer, I will measure claim-evidence support, citation precision and coverage, use of retired evidence, ambiguity disclosure, false-premise resistance, and human-rated usefulness on a stratified sample. Ev2R-style evidence evaluation will be combined with human annotation rather than used as a sole judge.

All principal comparisons will use pre-specified trajectory-level splits, bootstrap confidence intervals, and paired tests where assumptions are met. Results will be replicated across at least two model families and with open-weight models sufficient to reproduce the core experiments. Ablations, cost-matched comparisons, temporal-leakage checks, and a predefined error taxonomy will test whether improvements follow from the claimed mechanism.

Feasibility, ethics, and risk control

The core programme is intentionally limited to textual and time-series evidence; multimodal extensions are optional. A dataset of 1,200-2,000 claim-time states is large enough for transition analysis while remaining annotatable within a PhD. Archived public sources and open models prevent dependence on a commercial API or an unconfirmed professional partnership. The work will retain source licences and timestamps, minimise redistribution of copyrighted content, avoid unnecessary repetition of harmful claims, and document provenance and takedown procedures. Human annotation will receive ethics review where required, with annotators warned about potentially distressing misinformation. Source authority will not be treated as a universal scalar truth: experiments will expose provenance, model source dependence explicitly, and report sensitivity to alternative source-weighting assumptions.

Work plan and first-year deliverables

Months 1-6: complete a systematic task and prior-art audit; formalise the state and transition taxonomy; reproduce strong static, insufficient-evidence, and iterative-retrieval baselines; build an archived evidence pipeline; and annotate a pilot of at least 100 trajectories or 300-400 claim-time states.

Months 7-12: validate annotation quality; release a preregistered benchmark protocol and baseline suite; conduct the first analysis of commitment-state-communication failures; and submit a first paper centred on the task formulation and diagnostic findings. These are concrete first-year deliverables even if later modelling choices change.

Year 2: complete the longitudinal resource and develop calibrated sequential commitment policies, including cost- and risk-controlled evaluation.

Year 3: develop and evaluate revisable evidence states and state-constrained communication; integrate the components; conduct cross-domain and cross-model replication; and complete the substantive thesis contributions. A fourth year, if used, would be reserved for extensions, consolidation, and writing rather than essential data collection.

Research environment and preparation

The proposed direction is designed to align with several lines visible in Professor Andreas Vlachos's public research programme. AVeriTeC and Ev2R provide foundations for real-world evidence and its evaluation; AmbiFC and Dis2Dis expose claim ambiguity; TSVer and PledgeTracker show the importance of temporal and incremental evidence; and recent forecast-updating work raises related questions about calibrated updating under time-bounded information. The PhD would connect these lines through a general longitudinal verification problem and explicit state and transition evaluation. This fit is a hypothesis for discussion, not a claim that the project is distinct from unpublished work, that supervision is available, or that an agreement exists; the scope would be revised in response to a prospective supervisor's advice.

My Master's research provides methodological preparation rather than the PhD task itself. I designed MGP-6, an author-designed controlled diagnostic of persistent-memory decisions, state lineage, and answer realisation, and implemented controllers and external-framework baselines. On 180 scenarios, the LLM controller achieved 88.9% decision accuracy against 42.2% for naive overwrite, while independent annotation of the held-out split reached Cohen's kappa of 0.981. The principal lesson was that correct commitment decisions, correct persistent state, and correct answers are separable. The proposed PhD transfers that experimental discipline - not the personalised-memory action space - to public evidence verification. Its scientific objects are new: source-dependent external evidence, open-world acquisition, claim interpretation, temporal validity, and fact-checking verdicts. My additional work on ontology-guided representation learning, retrieval systems, and reproducible research software supports the structured-state and engineering demands of the programme.

Expected contributions

The thesis is expected to contribute: (1) a precise longitudinal task and reproducible resource for evolving-evidence verification; (2) calibrated methods for deciding when to search, defer, commit, and revise; (3) provenance-rich state representations and update rules with transition-level evaluation; (4) evidence establishing when state-constrained generation does or does not improve factual communication; and (5) an evaluation methodology that distinguishes commitment, state, and communication failures. Together, these contributions would move automated verification from retrieving evidence for a single answer towards maintaining defensible, revisable commitments as the information environment changes.

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